

```
def dijkstra(graph, source):
```

```
    n = len(graph)
```

```
    distance = [9999] * n
```

```
    visited = [False] * n
```

```
    distance[source] = 0
```

```
    for i in range(n):
```

```
        minimum = 9999
```

```
        u = -1
```

```
        for j in range(n):
```

```
            if not visited[j] and distance[j] < minimum:
```

```
                minimum = distance[j]
```

```
                u = j
```

```
        if u == -1:
```

```
            break
```

```
    visited[u] = True
```

```
    for v in range(n):
```

```
        if graph[u][v] != 0 and not visited[v]:
```

```
            new_distance = distance[u] + graph[u][v]
```

```
            if new_distance < distance[v]:
```

```
                distance[v] = new_distance
```

```
    return distance
```

```
graph = [
```

```
    [0, 4, 1, 0],
```

```
    [4, 0, 2, 5],
```

```
    [1, 2, 0, 8],
```

```
    [0, 5, 8, 0]
```

```
]
```

```
source = 0
```

```
distance = dijkstra(graph, source)
```

```
print("Shortest distances from source", source)
```

```
for i in range(len(graph)):
```

```
    print("Vertex", i, "=", distance[i])
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 19 2025)
on win32

Enter "help" below or click "Help" above

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/C
```

```
Shortest distances from source 0
```

```
Vertex 0 = 0
```

```
Vertex 1 = 3
```

```
Vertex 2 = 1
```

```
Vertex 3 = 8
```

```
Time Complexity = O(V^2)
```

```
>>>
```